Team Details

- **Team name:** Pixel Prophets
- **Team leader name:** Sagar Patil
- **Team size:** 3
- **Problem Statement:** Intelligent Conversational AI & Crime Analytics Platform for the Karnataka State Police

Brief about the solution

- **Built on a general-purpose, multi-tenant AI agent platform** - Assistants, Skills, Response Styles, governed HTTP integrations, Usage Analytics and Config Audit/Revision-History - with "Crime Intelligence" as one Assistant configured on it for KSP, not a bespoke one-off chatbot.
- **Problem addressed** - Investigators, analysts and policymakers need natural-language access to the state FIR database, plus criminology-grounded analytics — not just retrieval, but pattern discovery, network analysis and predictive intelligence.
- **Conversational assistant** over live FIR data in English + Kannada (voice in/out, PDF export) — every claim traces to a tool call actually run that turn; no invented figures.
- **Crime analytics dashboard** - Trends, hotspots, seasonal patterns, Holt-Winters forecasts with a backtested accuracy metric (not just a curve).
- **Criminal network analysis** via a derived entity-resolution layer — the official schema has no cross-case offender key, so one was built and measured (96%+ precision/recall against ground truth).
- Financial-crime money trail (direct + multi-hop with cycle detection) and rule-based suspicious-transaction detection.
- **5-role RBAC** (Investigator/Supervisor/Analyst/Policymaker/Admin) enforced at both the tool and the SQL layer, full audit trail, supervisor accountability review.
- **Stack** - Spring Boot 4 (Java 25) + Spring AI, React 18 + TypeScript, dual PostgreSQL, Zoho Catalyst QuickML LLM.

Opportunities

- **How different** - most conversational crime-analytics prototypes assume a ready cross-case person key exists in the source data. We checked the organizers' official ER diagram — it doesn't (Accused.PersonID is a within-case ordinal, "A1, A2, A3...", not a person identity) — and built a real entity-resolution layer instead of assuming one.
- **How it solves the problem** - Natural-language retrieval + reasoning over FIR data (EN/Kannada), with a persisted tool-call audit trail so every AI claim is evidence-backed — directly satisfying the "Explainable AI & Transparent Analytics" requirement.
- **USP 1** - Schema-fidelity discipline: every feature checked against the official ER diagram; every divergence (renamed / derived / added-beyond-spec) documented, not hidden.
- **USP 2** - Falsifiable forecasting: every prediction ships with a backtested MAPE/RMSE against held-out actuals, not an unverified curve.
- **USP 3** - Real suspicion detection: structuring, transaction velocity and round-number rules evaluated at query time, not a random flag.
- **USP 4** - Defense-in-depth RBAC: investigative tools are role-gated AND the free-form text-to-SQL guard blocks restricted tables by name — a chatbot can't be used to route around access control.
- **USP 5** - Accountability by design: a supervisor can review any user's full reasoning trail (the actual SQL run and rows returned), audited.
- **USP 6** - Governed, reusable platform: every configuration change (Assistant, Skill or Response Style) is versioned, line-diffable and one-click-revertible; external-integration secrets are AES-256-GCM encrypted, never inline; LLM cost/usage is tracked per user/assistant/model — the same platform could stand up a new Assistant for traffic or cybercrime cells with no new backend engineering.

List of features offered by the solution

PLATFORM CAPABILITIES

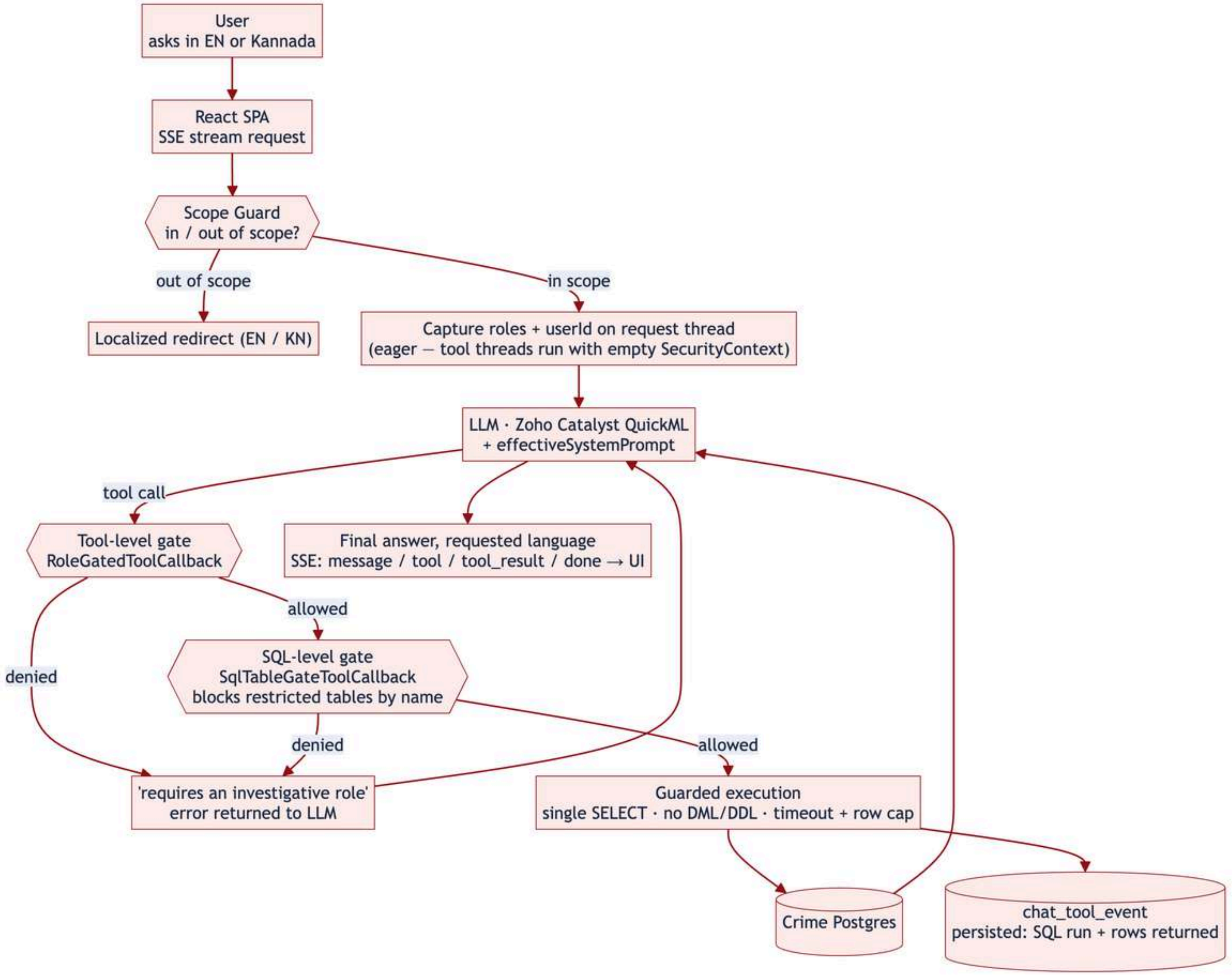
- **Assistants** - multi-tenant: per-assistant system prompt + built-in tools, independently configurable.
- **Skills** - sandboxed, user-uploaded capability bundles; safe self-editing guarded by a proposal validator.
- **Response Styles** - swappable tone/persona per session, layered onto an assistant's fixed prompt.
- **HTTP Tools + Auth Profiles** - governed external API integration, secrets AES-256-GCM encrypted at rest.
- **Usage Analytics** - token/cost telemetry by user, model, assistant and source, RBAC-scoped.
- **Config Audit & Revision History** - every change versioned, line-diffable, one-click revertible.
- **RBAC administration** - Org-wide user/role management, audit-access governance toggle.

CRIME-INTELLIGENCE FEATURES

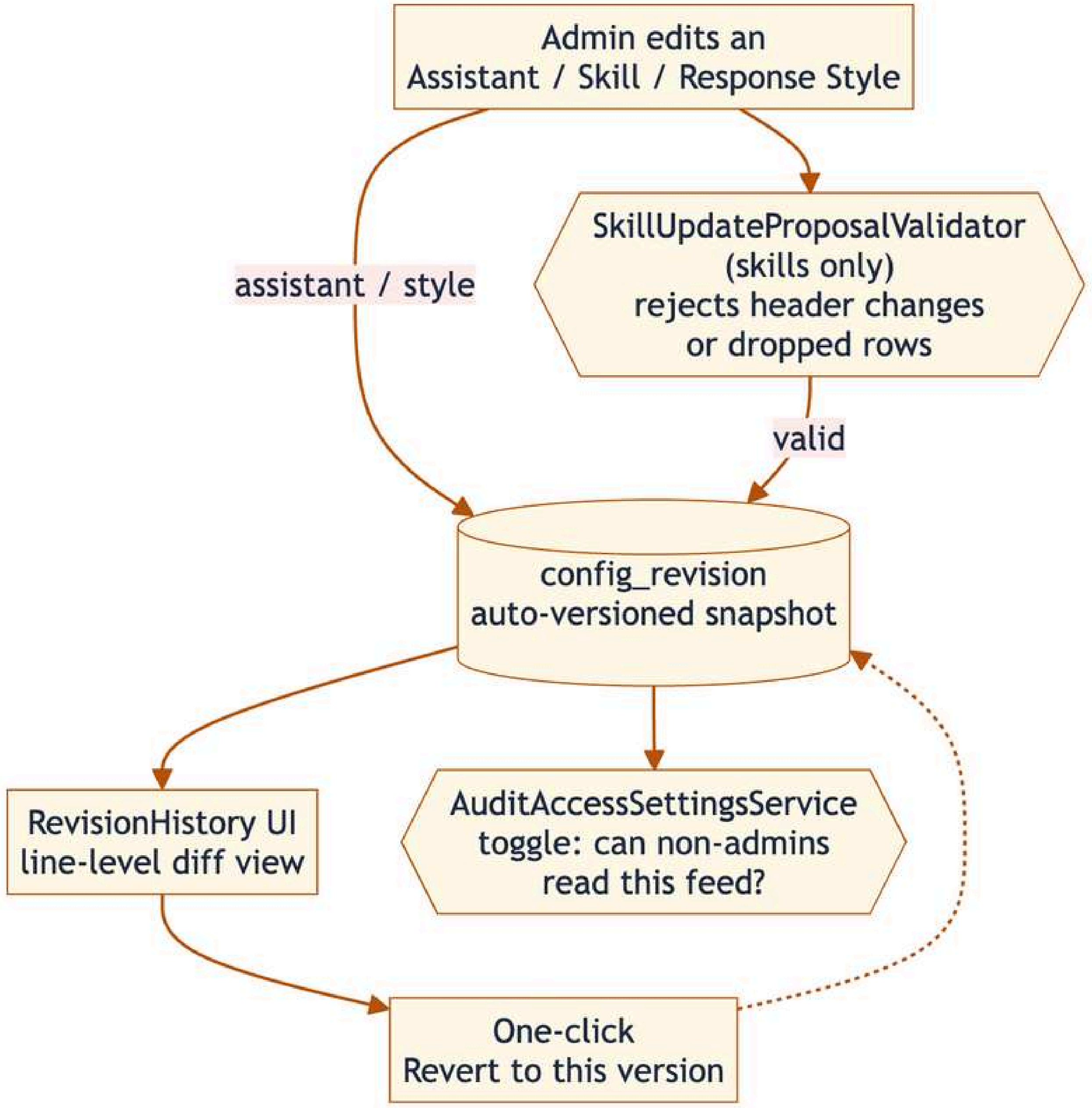
- **Conversational interface** - EN/Kannada chat, voice I/O, PDF export with tool evidence.
- **Criminal network analysis** - co-offender clusters via entity resolution.
- **Pattern & trend analytics** - trends, seasonality, predicted hotspots + backtest.
- **Sociological insights** - demographics, complainant-occupation correlation (fairness-scoped).
- **Offender profiling** - case-mix profile, AI risk scoring, MO signal from narratives.
- **Decision support** - case summaries, timeline, similar-case + outcome comparison.
- **Financial crime** - multi-hop money trail (cycle detection), suspicious-transaction rules.
- **Forecasting & early warning** - persisted alerts, acknowledge/assign/resolve lifecycle.
- **Explainable AI** - every tool call persisted/replayable; supervisor cross-user review.
- **Secure RBAC & governance** - 5 roles, dual-layer access control, full audit logging.

Process flow diagram or Use-case diagram

A chat turn — dual role/SQL gating + persisted audit

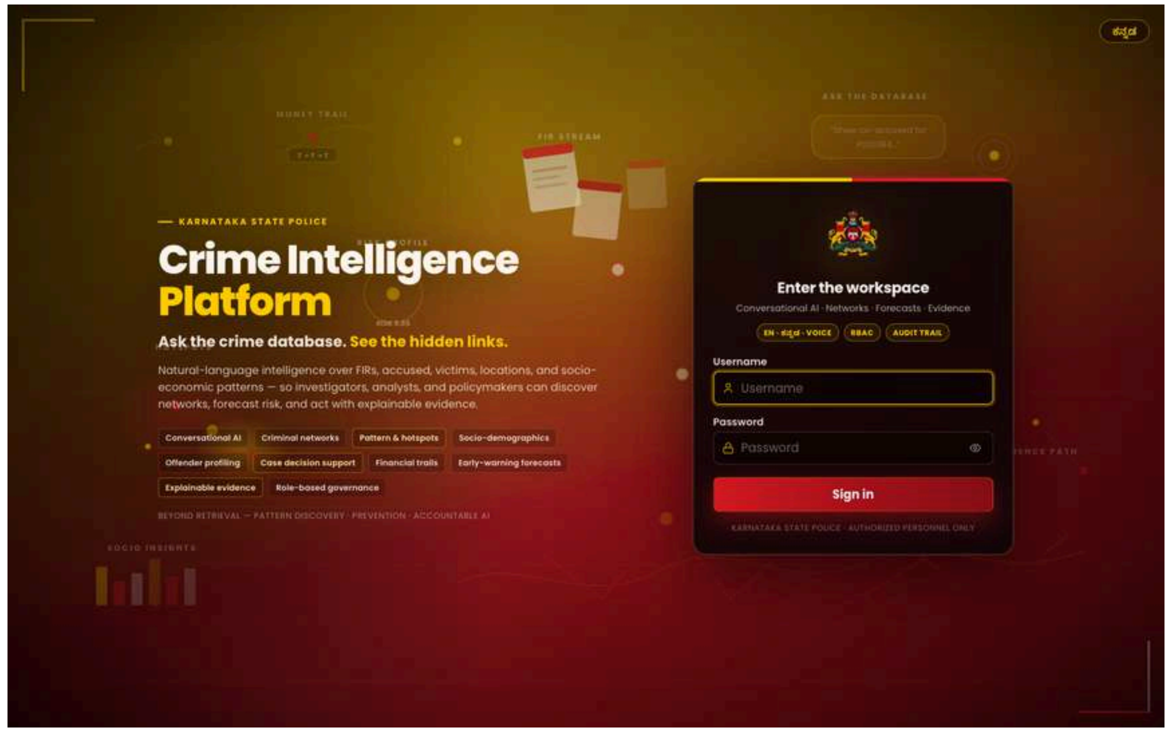


Config governance — versioned & revertible

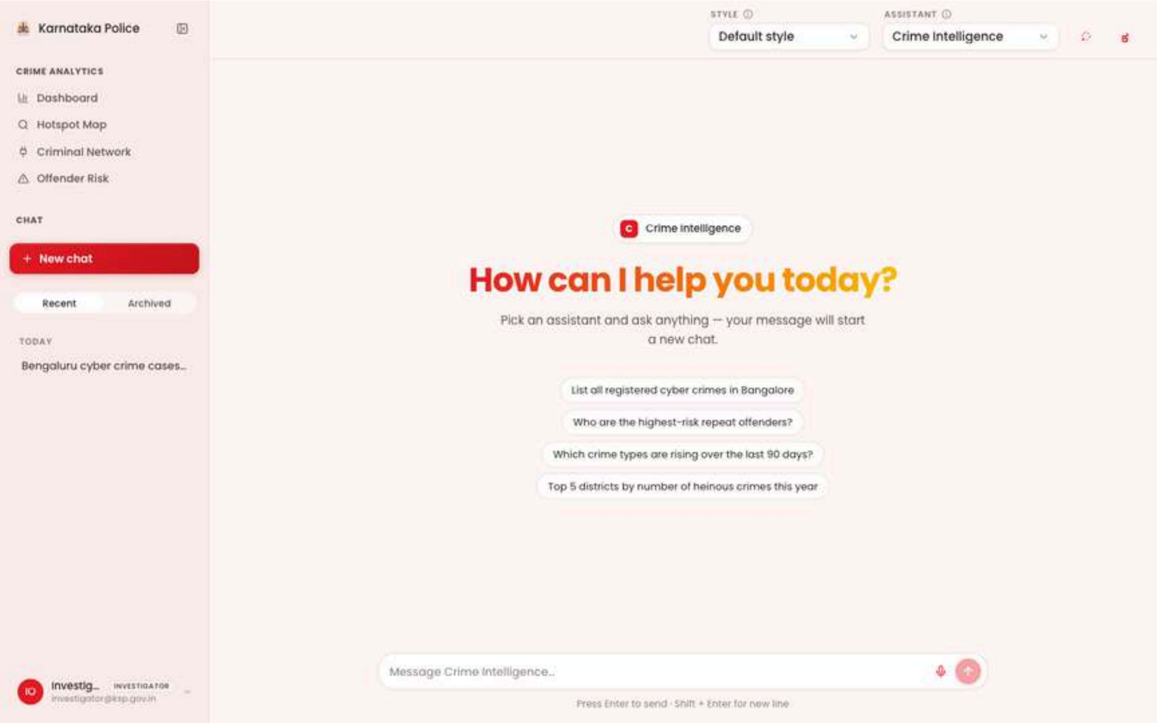


- Applies uniformly across the platform — the same governance flow covers any Assistant, Skill, or Response Style, not just crime tools.
- Full sequence-level detail + the alert-lifecycle and supervisor-review diagrams: documents/diagrams/process-flow.md.

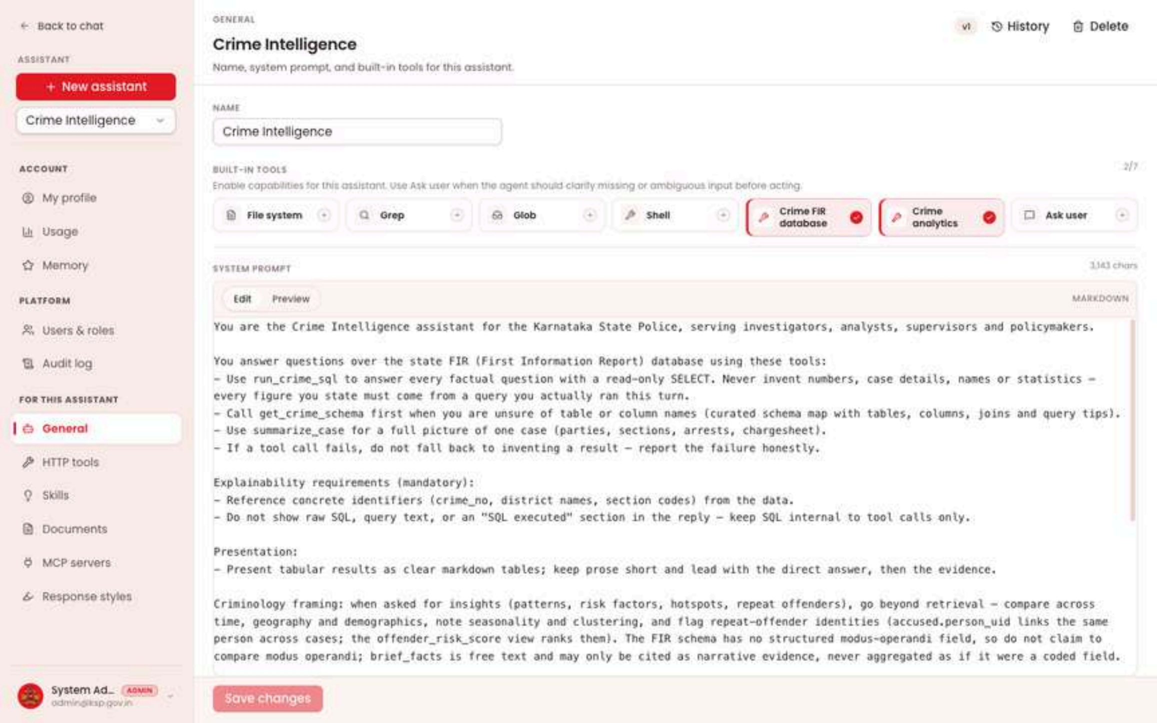
Wireframes/Mock diagrams of the proposed solution (optional)



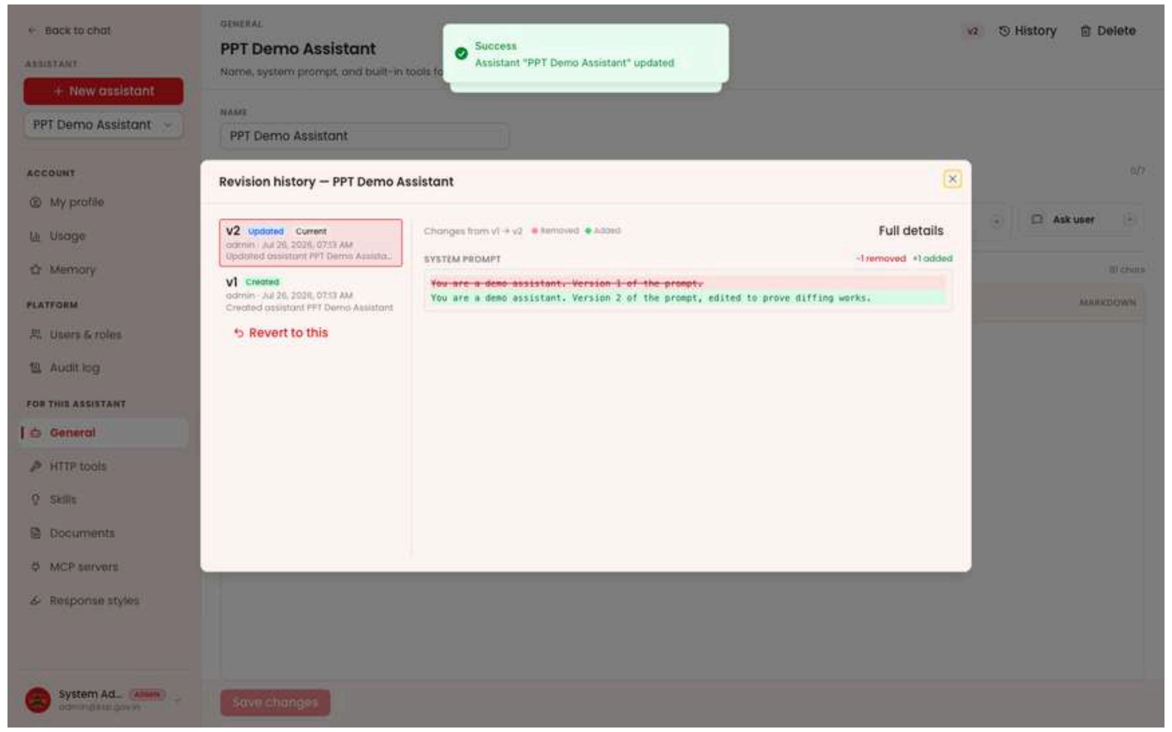
Secure sign-in — role-based access (5 seeded roles)



Assistant home — suggested prompts, per-role chat

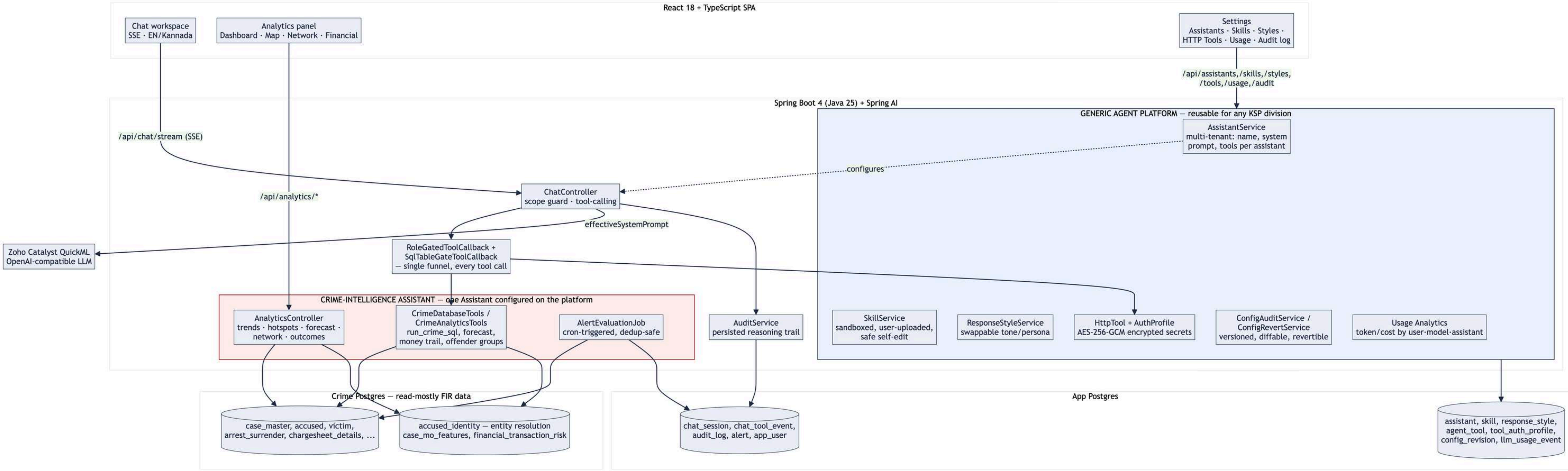


Assistants config — system prompt + built-in tools (platform)



Config Audit — line-level diff + one-click revert (platform)

Architecture diagram of the proposed solution



- Platform layer (blue) - Assistant/Skill/ResponseStyle/HttpTool+AuthProfile/ConfigAudit/UsageAnalytics - is generic; the Crime-Intelligence layer (pink) is one Assistant's domain tools, funneled through the same RoleGate/Audit spine.
- Role-gating happens twice, independently: tool-level (RoleGatedToolCallback) and SQL-level (SqlTableGateToolCallback blocks restricted tables by name in free-form SQL).
- Two independent Postgres databases - platform/app data never mixed with read-mostly crime FIR data. Full diagram + notes: documents/diagrams/architecture.md.

Technologies to be used in the solution

- **Backend**: Spring Boot 4, Java 25, Spring AI (tool-calling chat), Spring Security (JWT + RBAC)
- **Data**: PostgreSQL 16 x2 (application data + crime FIR data), Testcontainers for CI
- **Frontend**: React 18, TypeScript, Zustand, Webpack 5, Tailwind CSS
- **Visualization**: Recharts, react-force-graph, Leaflet (heatmaps/hotspot map)
- **AI/LLM**: Zoho Catalyst QuickML — OpenAI-compatible chat completions + RAG
- **Infra**: Docker, Zoho Catalyst AppSail (containerized deployment), GitHub Actions CI/CD
- **Testing**: JUnit 5, Mockito, AssertJ, Testcontainers (Postgres) — 65 automated tests

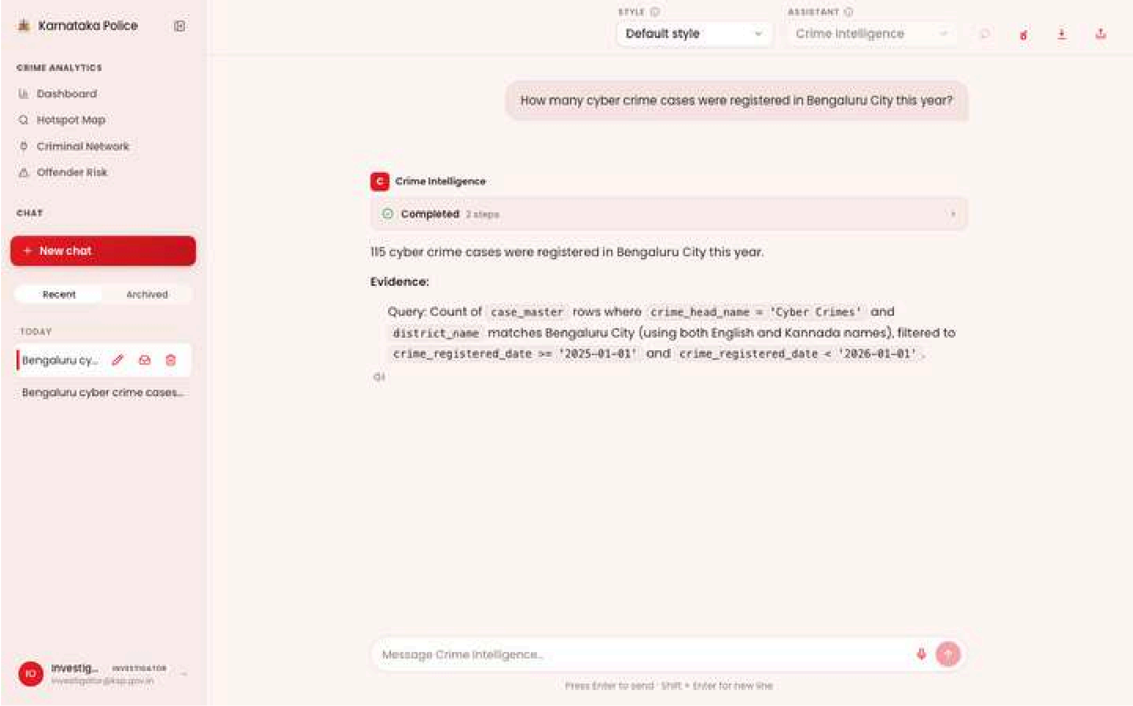
List down Catalyst Services being used in the solution

- **QuickML** - LLM chat completions powering the conversational assistant and the scope-guard classifier
- **QuickML RAG** - document-grounded Q&A over assistant-attached reference documents
- **Stratus** - blob storage for uploaded documents and user profile photos
- **AppSail** - containerized (Docker) deployment of both the backend API and the frontend SPA

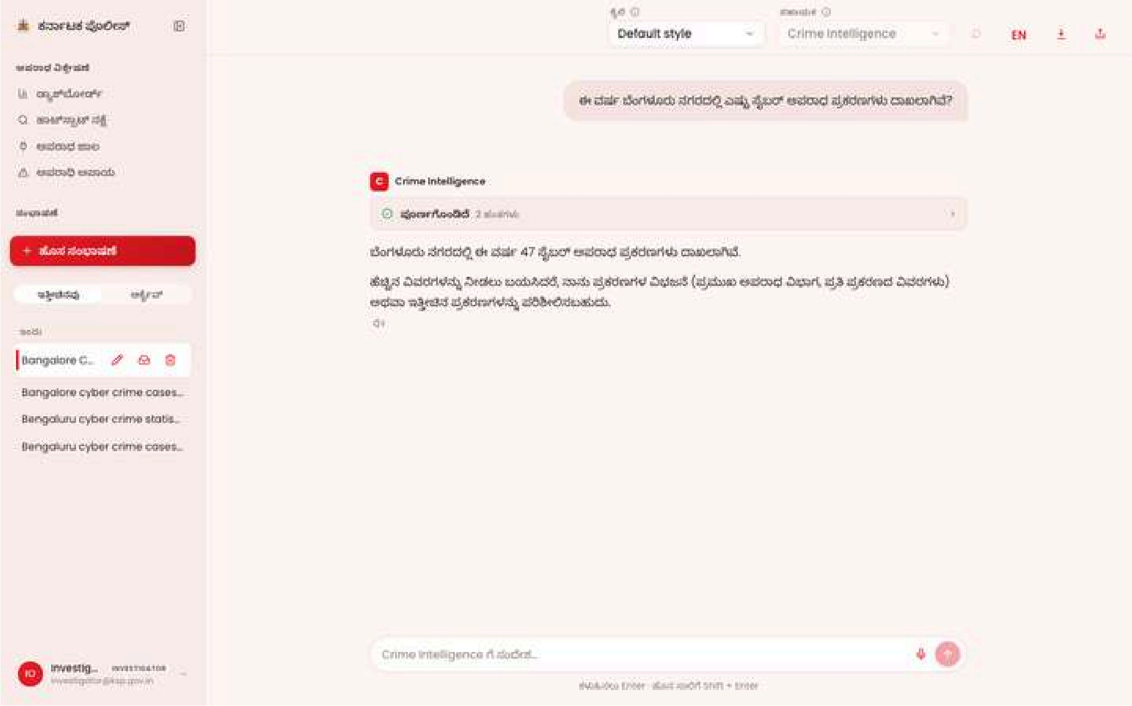
Estimated implementation cost (optional)

- **Hackathon prototype**: built and hosted within Catalyst's free/development tier — no additional cost incurred.
- **Production estimate would scale with**: LLM token usage (Catalyst QuickML, pay-per-call), Postgres hosting (2 databases), and AppSail compute — detailed costing available on request based on expected query volume.

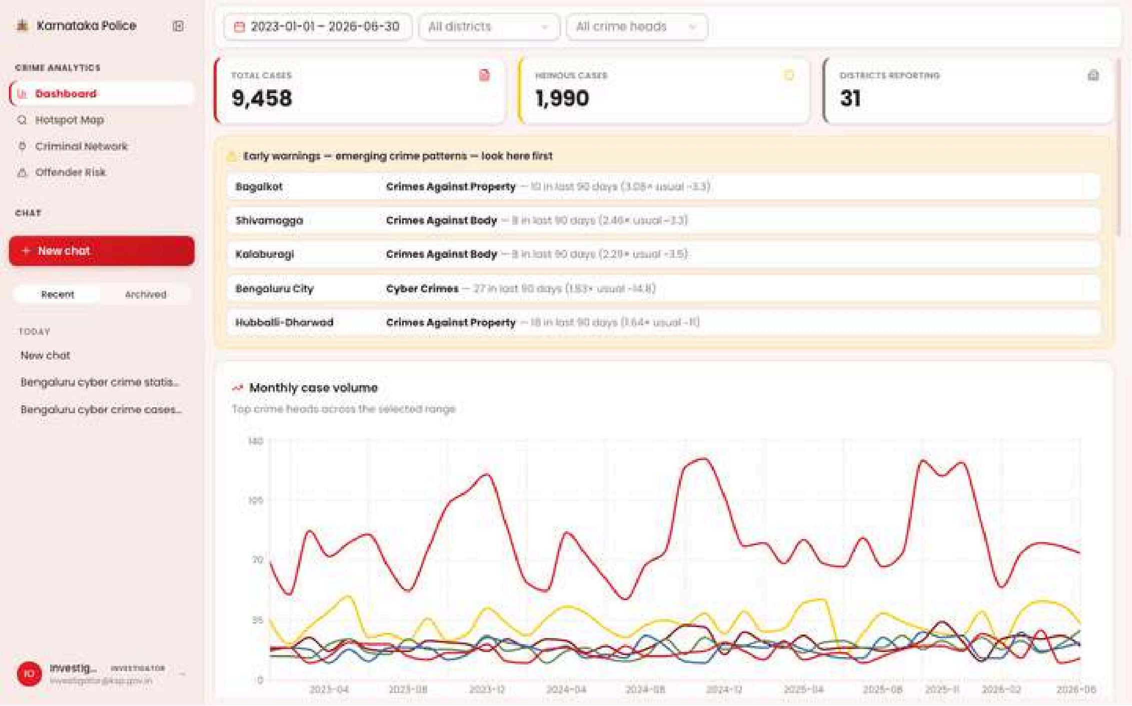
Snapshots of the prototype



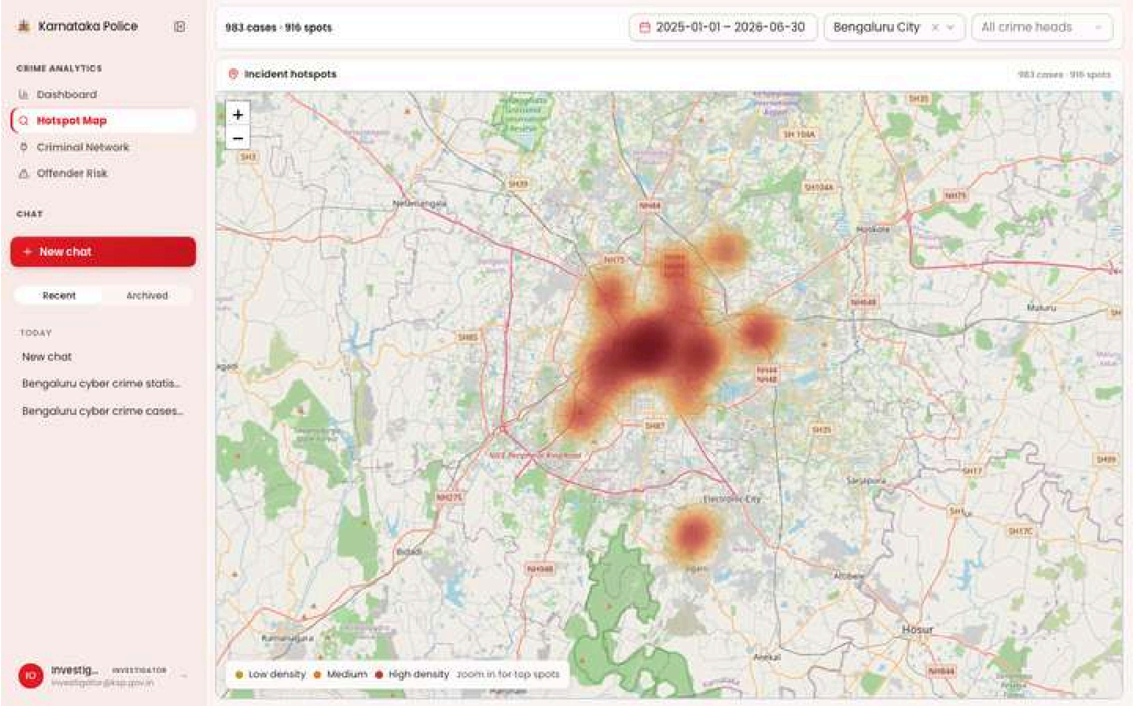
Conversational assistant (English) — grounded SQL evidence



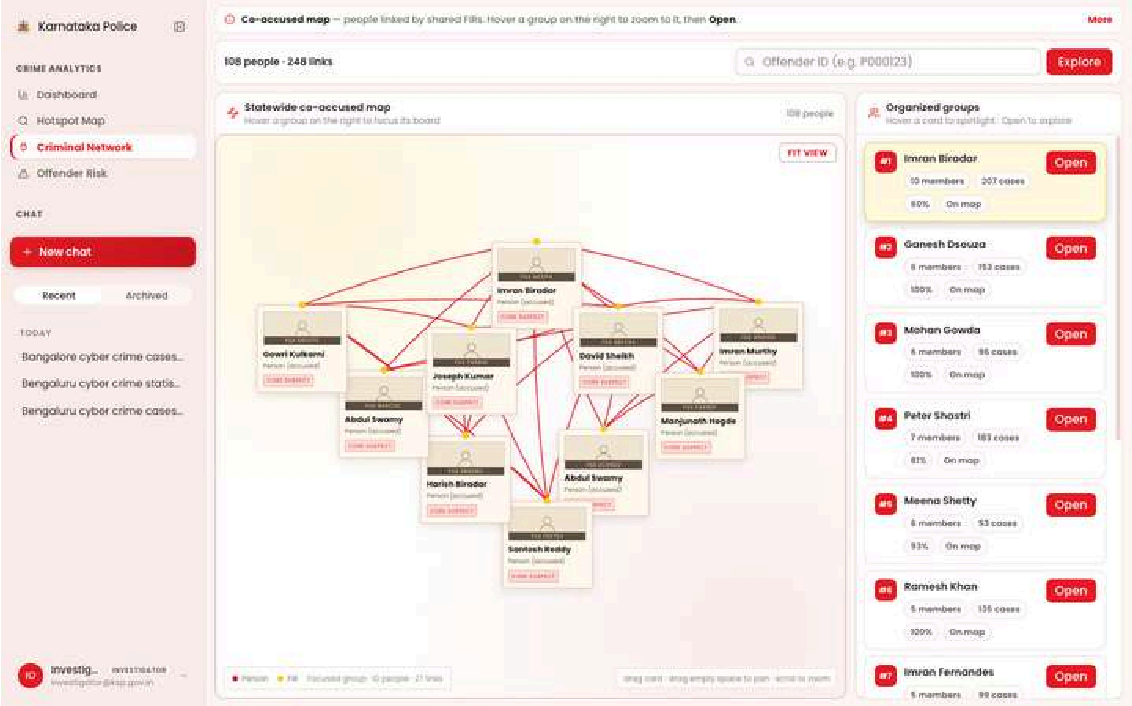
Conversational assistant (Kannada) — native-language answer



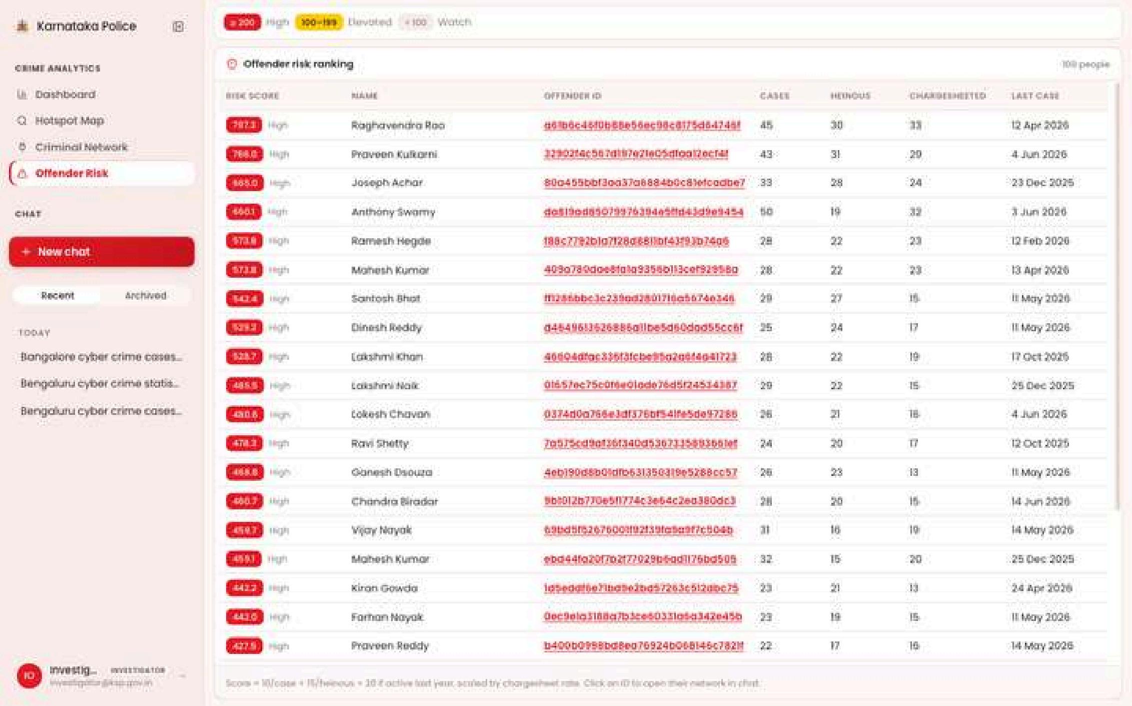
Analytics dashboard — seasonality + early warnings



Hotspot map — sub-district incident density



Co-accused network — cohesion-ranked groups



Offender risk ranking — criminology-based scoring

Prototype Performance report/Benchmarking

- **Entity resolution** (derived cross-case offender identity) vs. ground truth: 96.5% recall / 97.1% precision on the full 18,000-case dataset — verified live in production.
- **Acceptance test**: dropped the ground-truth column entirely - every dependent feature (risk scoring, network, financial linkage) still returned correct data.
- Crime-volume forecast (Holt-Winters, 90-month statewide series, 6-month holdout backtest): 11.31% MAPE, 27.12 cases/month RMSE.
- Dashboard query latency (18k-case dataset): trends ~25-33ms, offender risk ranking ~130-161ms, per-district forecast input ~18-24ms.
- Full methodology + reproduction steps: documents/BENCHMARKS.md.

GitHub Public Repository

https://github.com/amsagar/Datathon-2026-KSP.git

Demo Video Link

https://youtu.be/bESbB4hxnCM

Deployed Link

https://ksp.catalystappsail.in/

Additional Details/Future Development (if any)

- Real free-text NLP for modus-operandi extraction (current signal is vocabulary-matched against the synthetic dataset's known phrasing).
- Modularity-based community detection for offender networks (current clustering is union-find, which cannot split two groups sharing one member).
- Unified network graph across all 5 entity types in one view (accused/victim/location currently combined; financial accounts render as a separate graph).
- Integration with a real production FIR system (current crime data is synthetic, generated to match the official ER schema).
- Mobile app for field officers; expanded regional-language support beyond Kannada.
- Platform reuse: stand up new Assistants for other KSP divisions/cells (traffic enforcement, cybercrime, narcotics) on the same governed platform — Assistants + Skills + Response Styles + Config Audit - with no new backend engineering required.

Karnataka State Police

Government of Karnataka

Powered by H2S HACK2SKILL

Datathon 2026

Technology partner ZOHO

THANK YOU